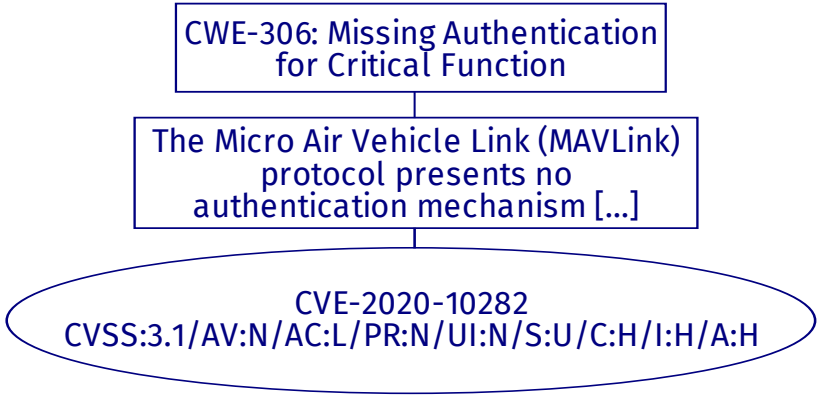
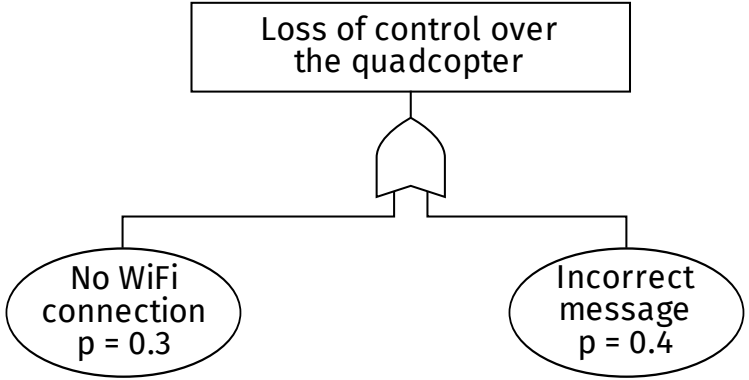
Combined Safety and Security Analysis in Self-Adaptive Systems

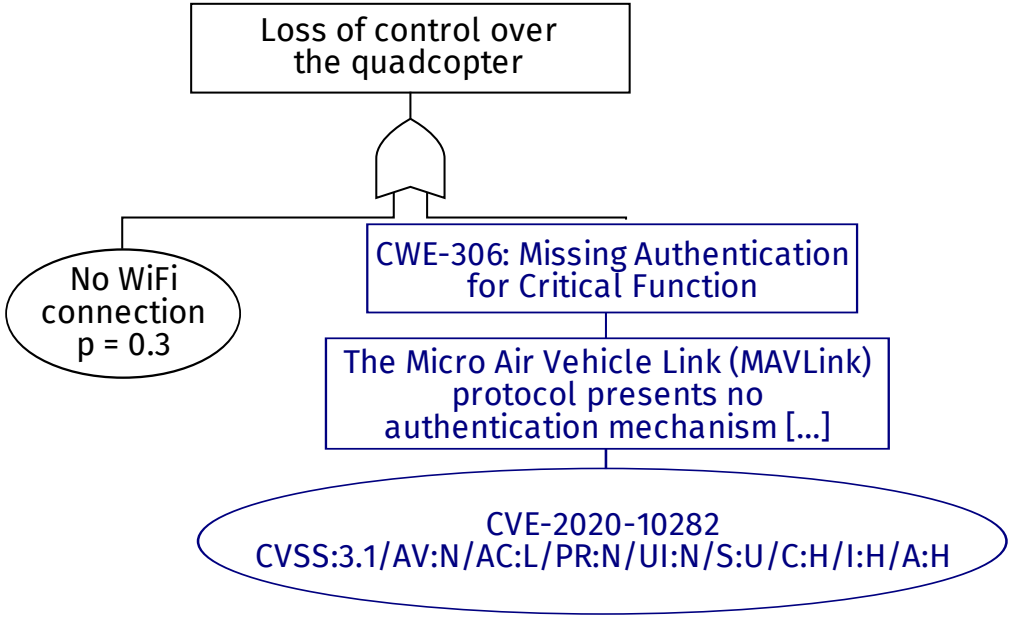
SafeSec Workshop – AFT Model Combination and Analysis

Safety & Security – Attack Fault Tree

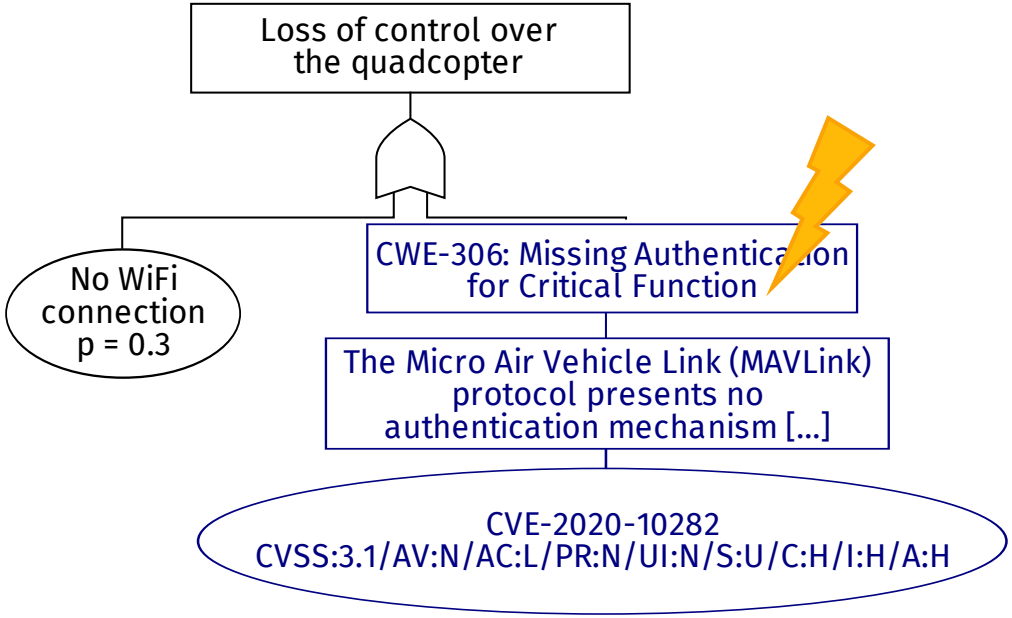




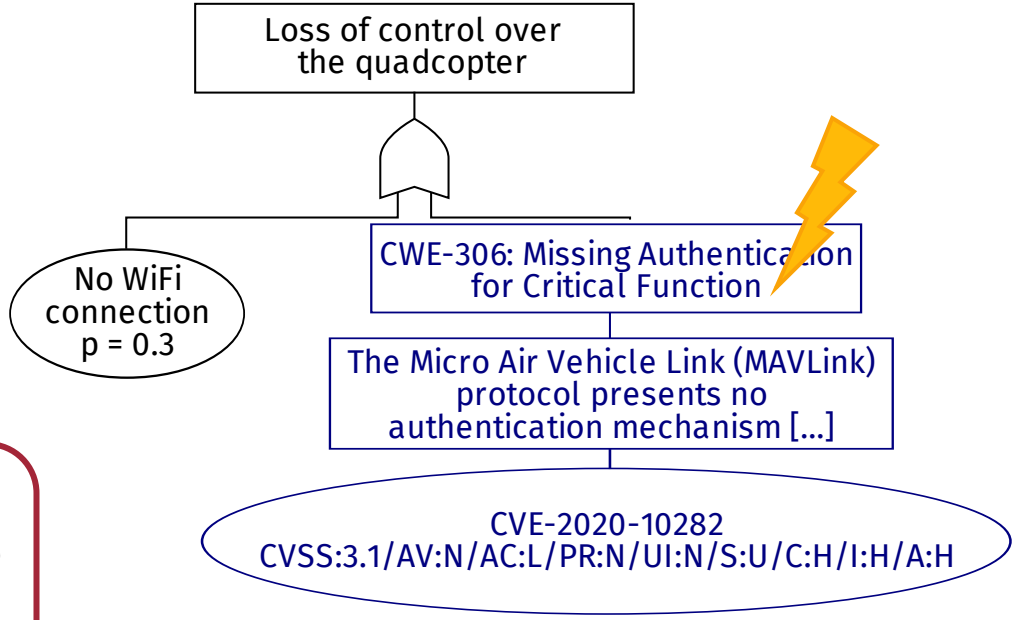
Safety & Security – Attack Fault Tree



Safety & Security – Attack Fault Tree



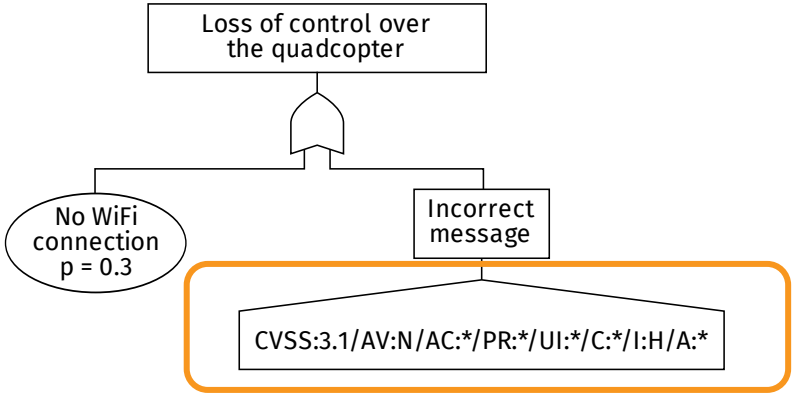
Safety & Security – Attack Fault Tree



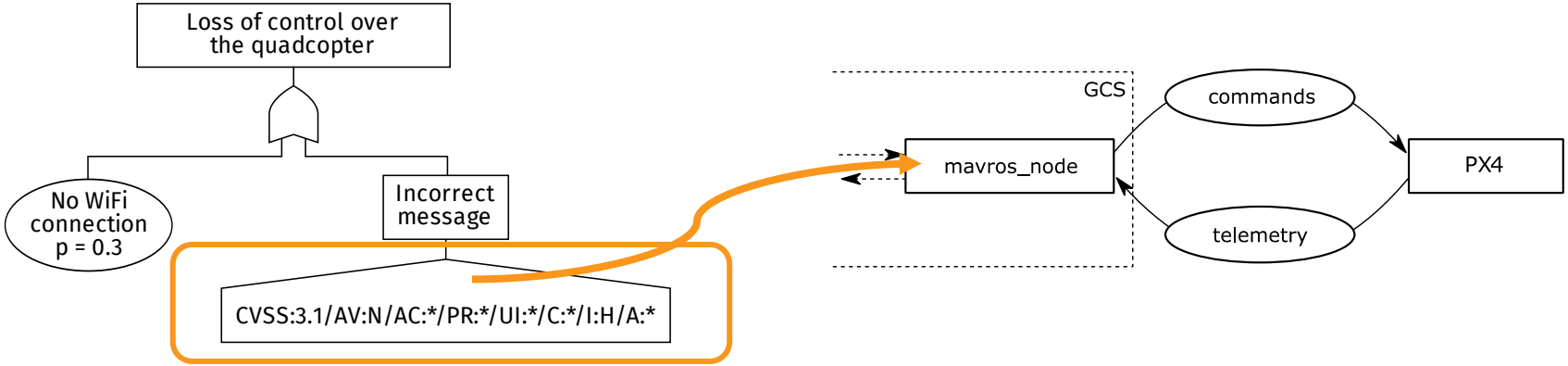
1. How to decide which Attack Tree should be attached to which part in the Fault Tree?

2. How to overcome different levels of abstraction?

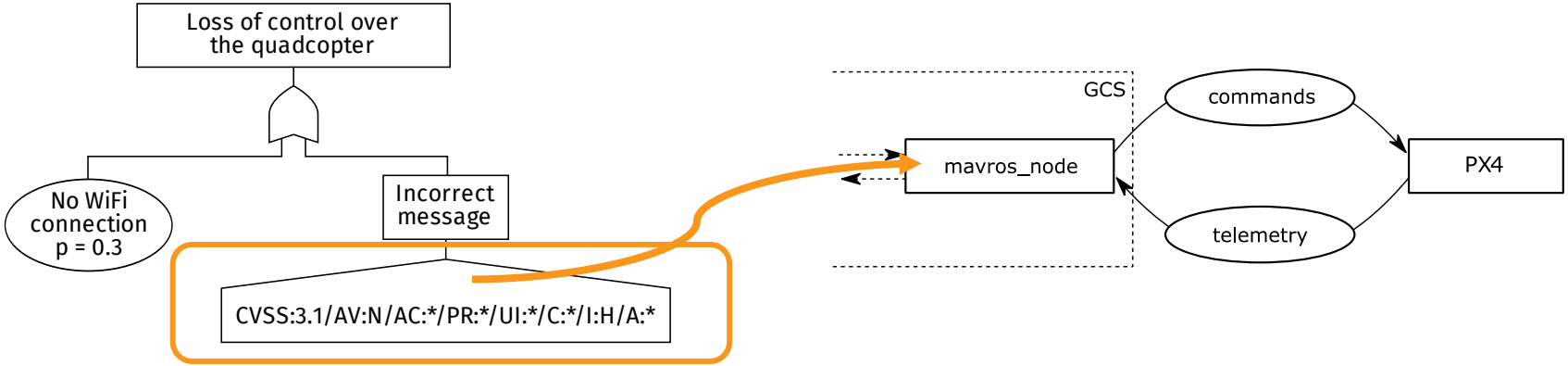
Safety – Fault Tree Extension



Safety – Fault Tree Extension



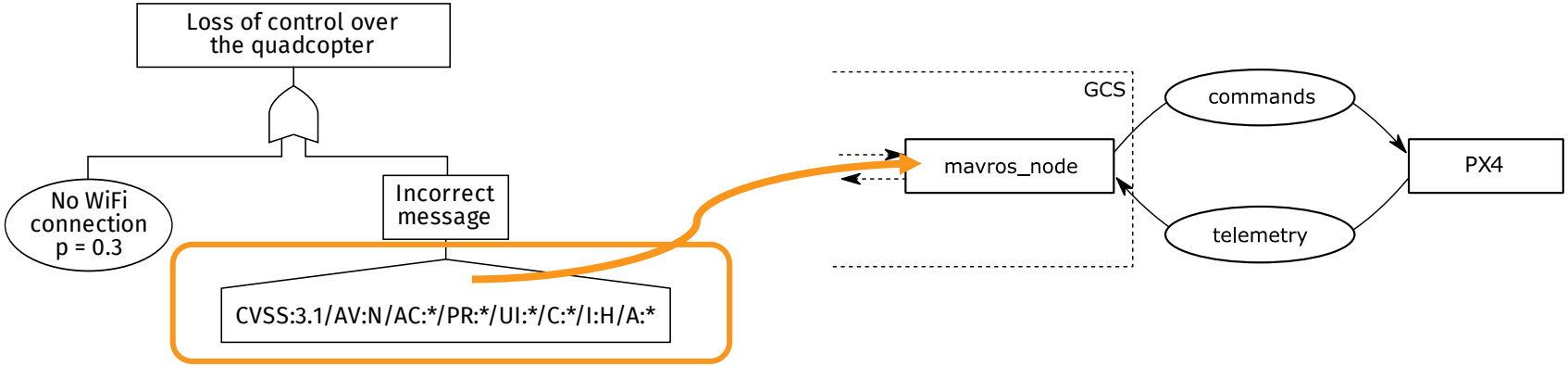
Safety – Fault Tree Extension



1. How to decide which Attack Tree should be attached to which part in the Fault Tree?

2. How to overcome different levels of abstraction?

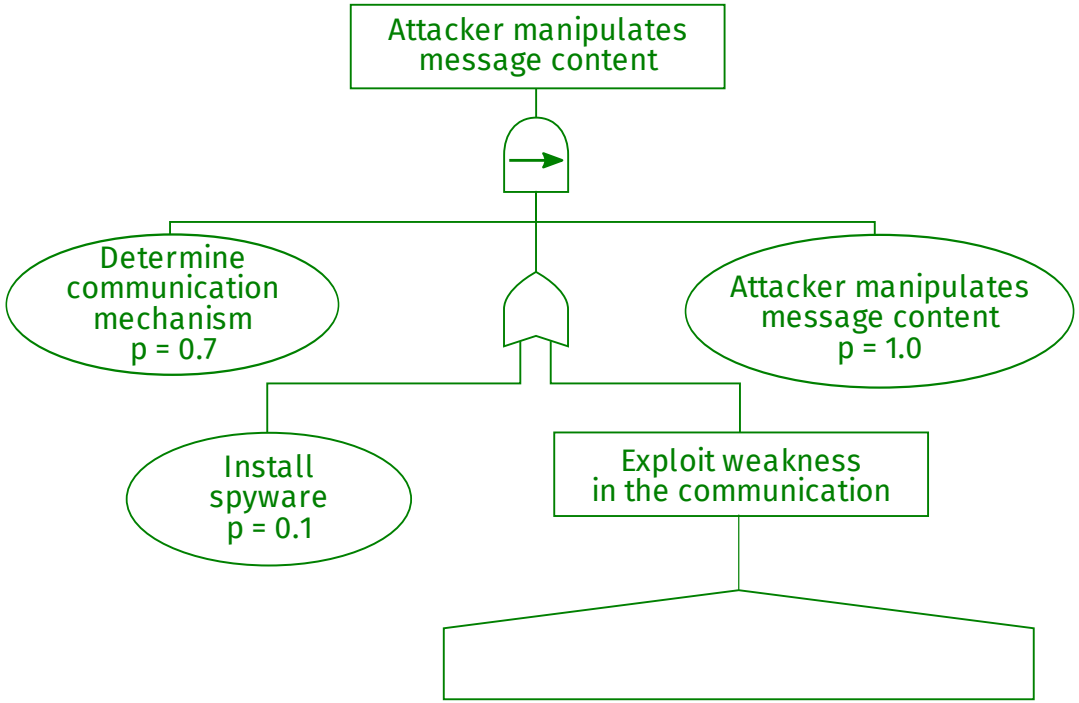
Safety – Fault Tree Extension



1. How to decide which Attack Tree should be attached to which part in the Fault Tree?

2. How to overcome different levels of abstraction?

Safety & Security – Attack Fault Tree Fragments

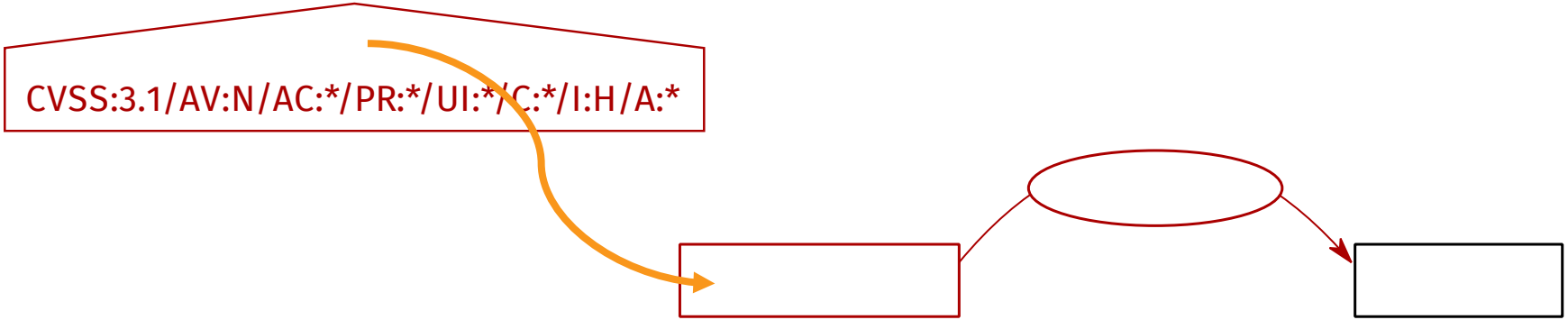


[1] https://capec.mitre.org/data/definitions/94.html

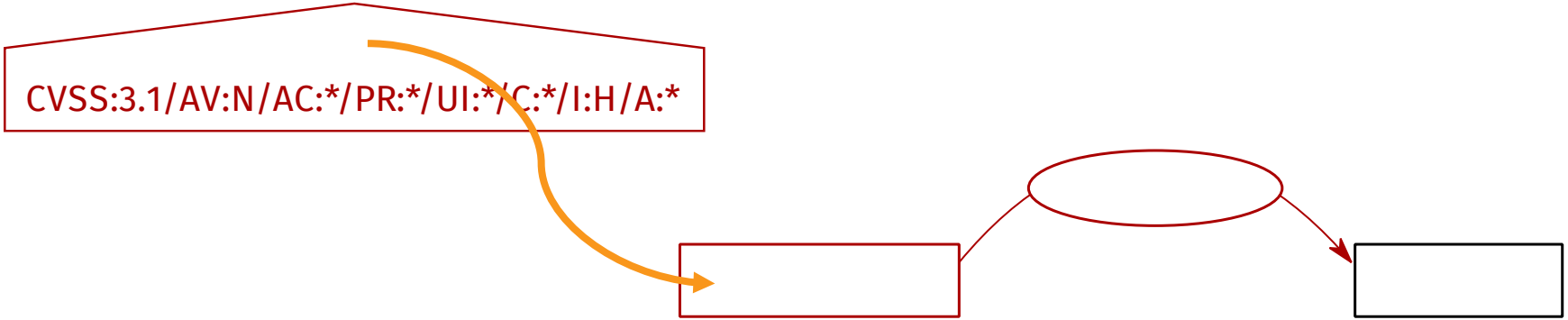
Safety & Security – Attack Fault Tree Fragments

CVSS:3.1/AV:N/AC:*/PR:*/UI:*/C:*/I:H/A:*

Safety & Security – Attack Fault Tree Fragments



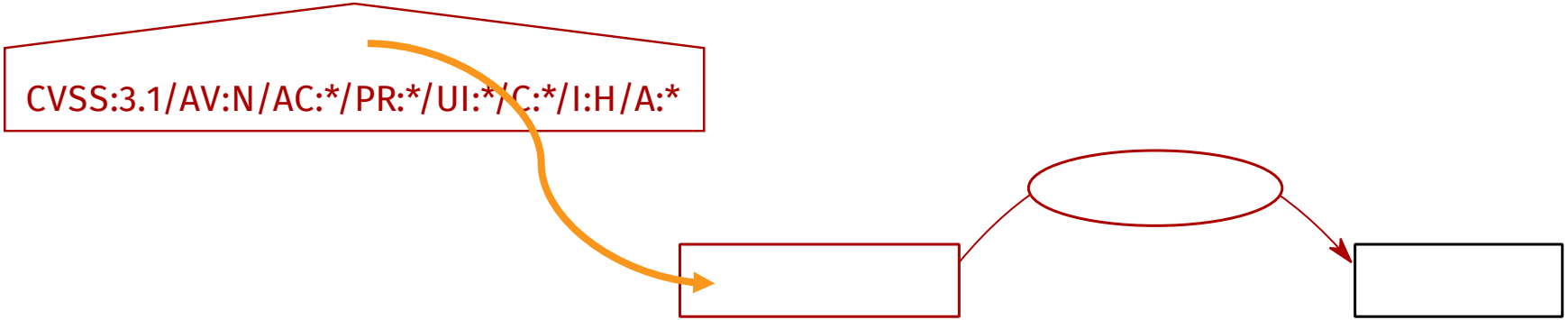
Safety & Security – Attack Fault Tree Fragments



1. How to decide which Attack Tree should be attached to which part in the Fault Tree?

2. How to overcome different levels of abstraction?

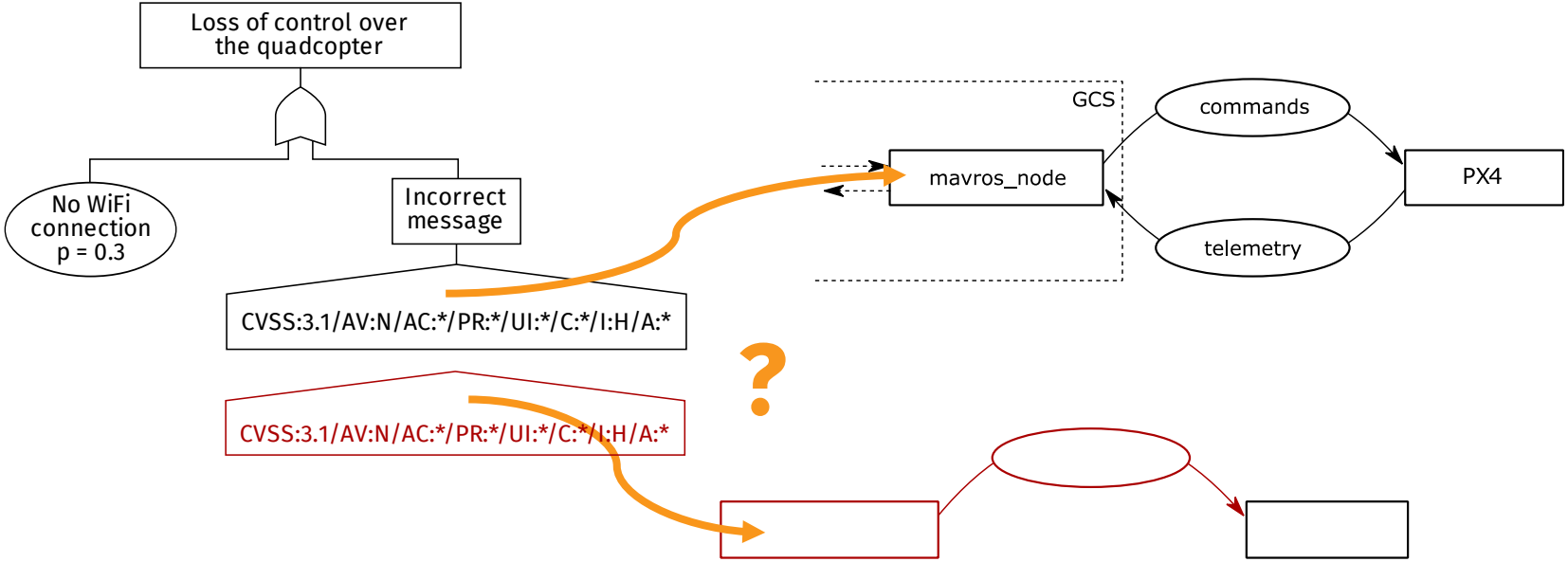
Safety & Security – Attack Fault Tree Fragments



1. How to decide which Attack Tree should be attached to which part in the Fault Tree?

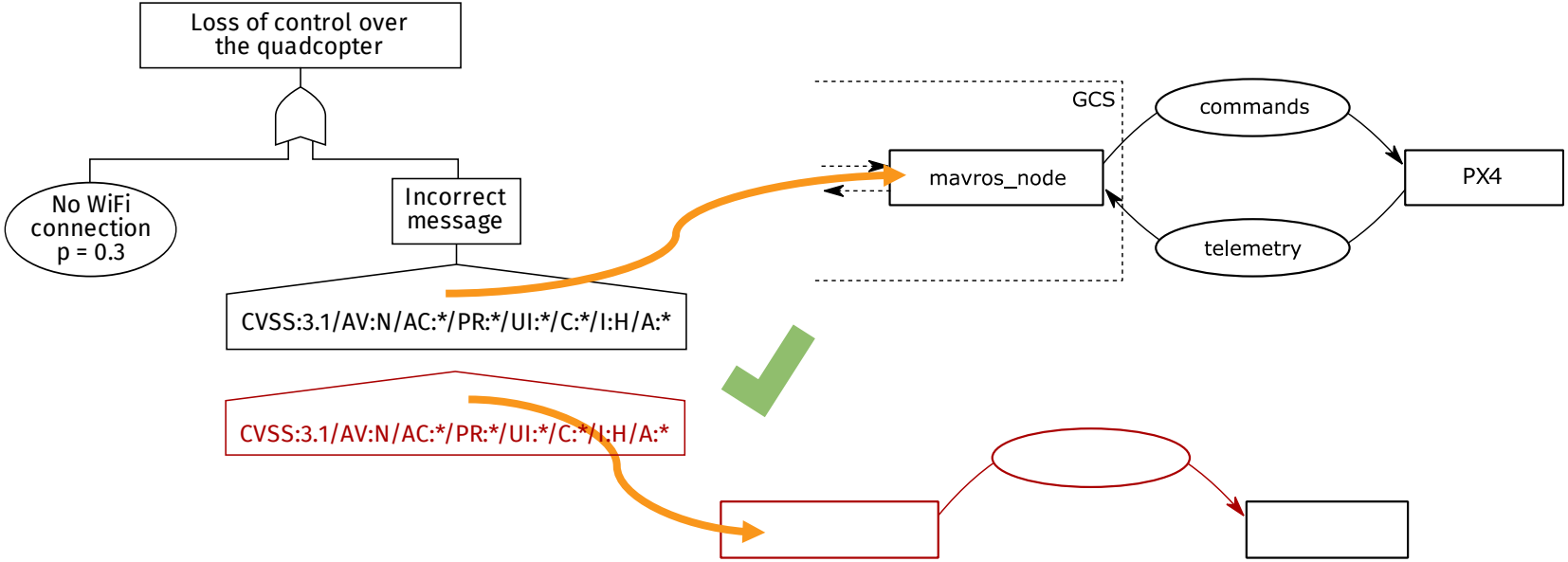
2. How to overcome different levels of abstraction?

Safety & Security – Attack Fault Tree Generator



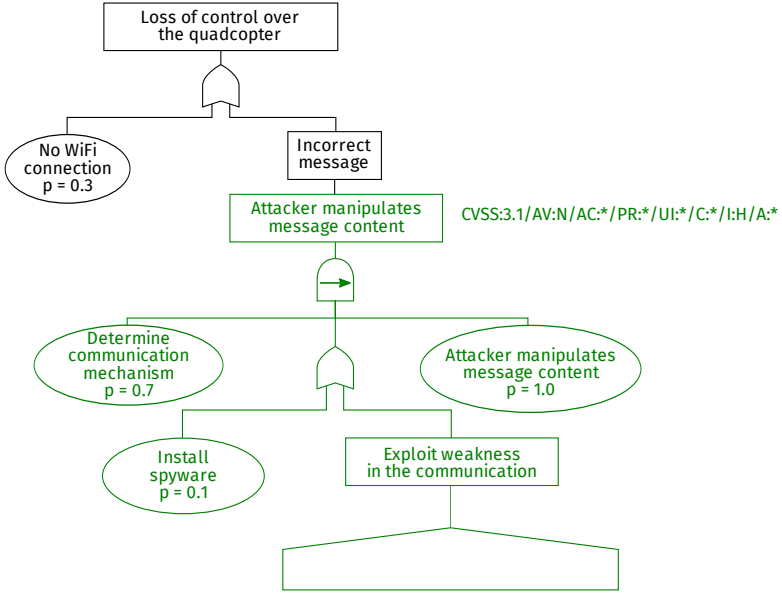
Fault Tree | AFT Fragments

Safety & Security – Attack Fault Tree Generator



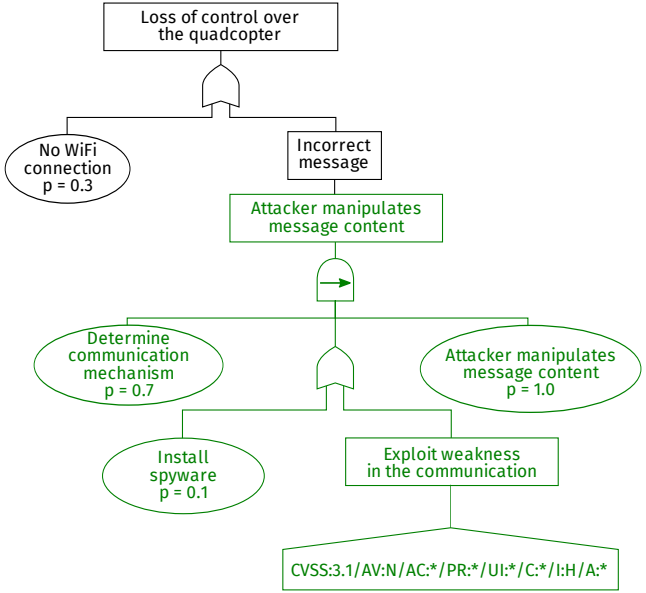
Fault Tree | AFT Fragments

Safety & Security – Attack Fault Tree Generator



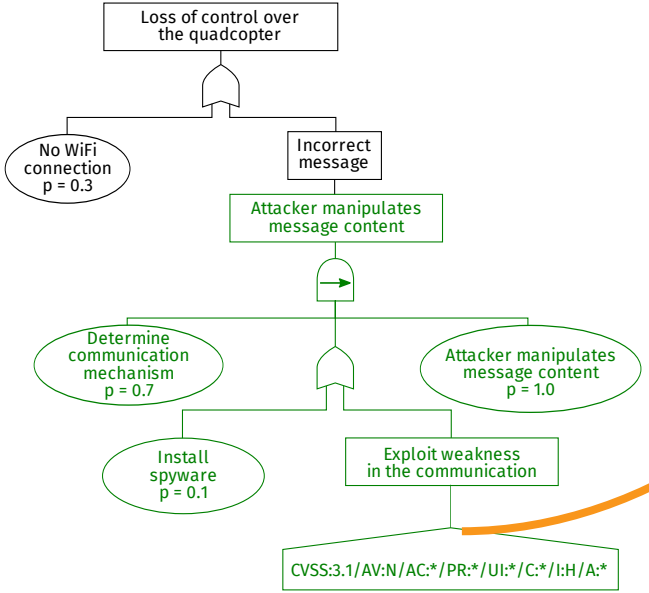
Fault Tree | AFT Fragments

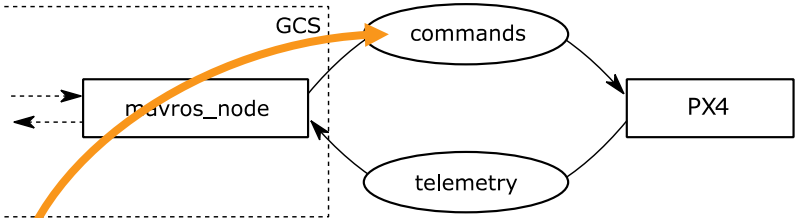
Safety & Security – Attack Fault Tree Generator



Fault Tree | AFT Fragments

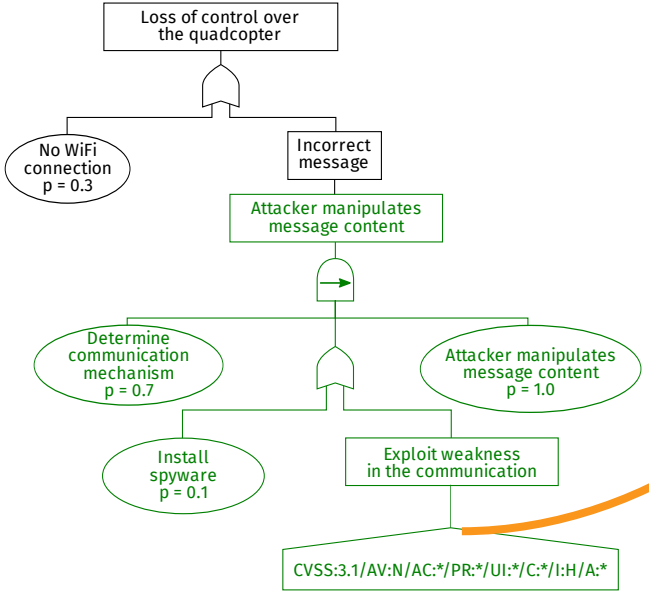
Safety & Security – Attack Fault Tree Generator

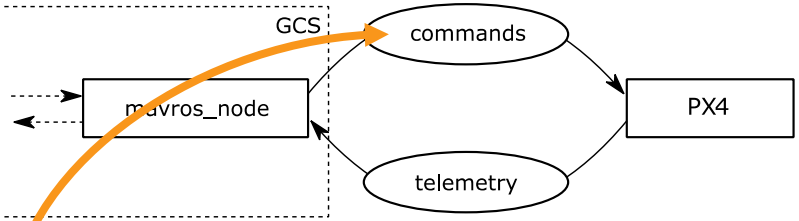




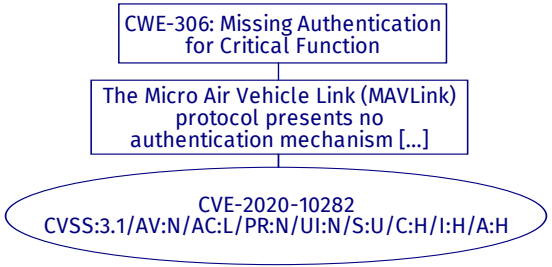
Fault Tree | AFT Fragments

Safety & Security – Attack Fault Tree Generator



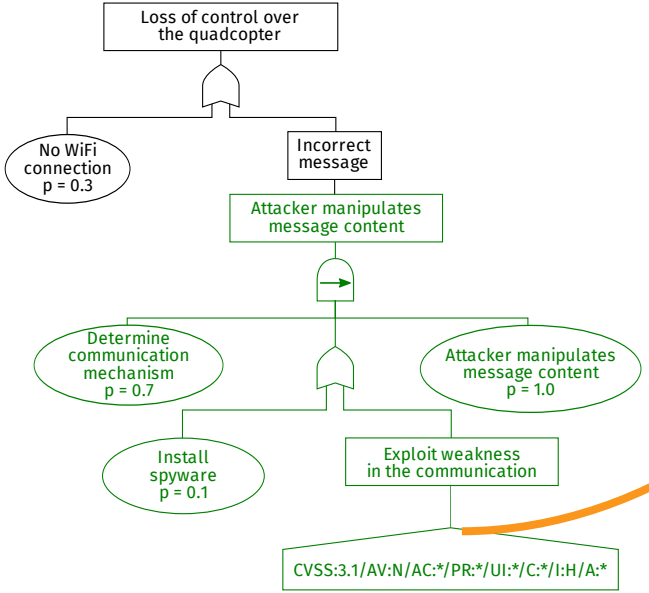


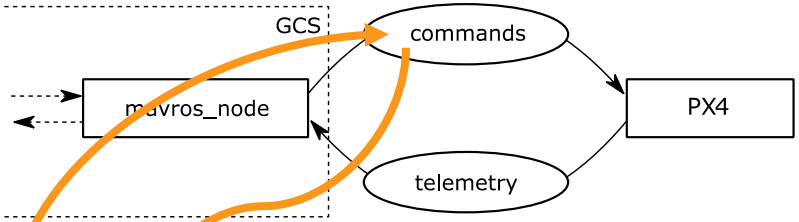
?



Fault Tree → AFT Fragments → Attack Fault Tree

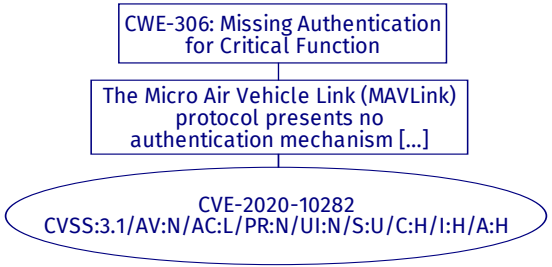
Safety & Security – Attack Fault Tree Generator





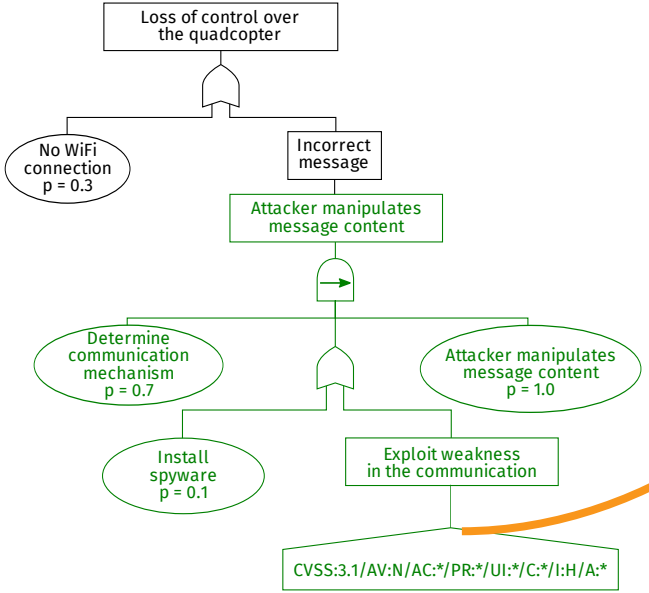

```
PX4:Component = {Mavlink1, pixhawk, ...}
Mavlink1:Protocol = {MavlinkLib, UDP, UART, ...}
commands:Channel = {Mavlink1, WiFi}
WiFi:Platform = {IEEE 802.11n, WPA2, UDP, TCP/IP, ...}
Radio:Platform = {UART}
mavros_node = {. . .} . . .
```

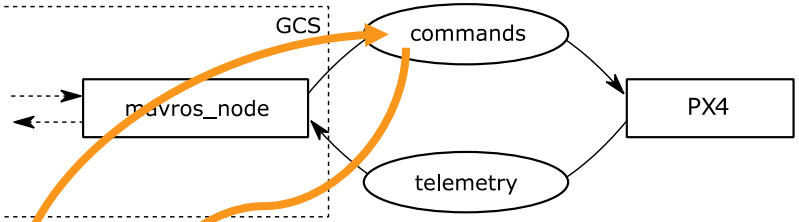
?



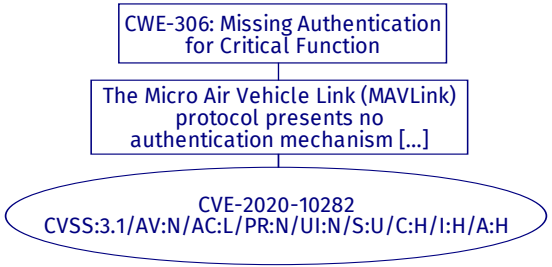
Fault Tree → AFT Fragments → Attack Fault Tree

Safety & Security – Attack Fault Tree Generator



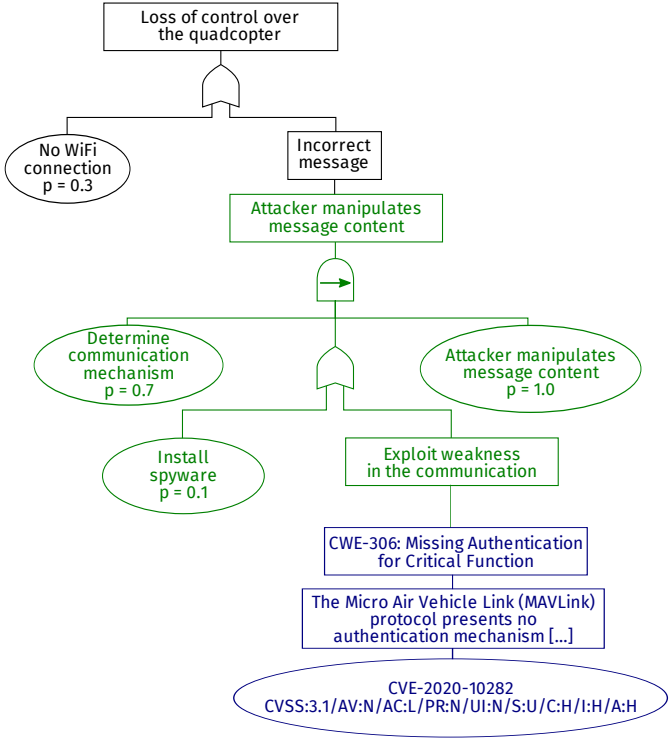



```
PX4:Component = {Mavlink1, pixhawk, ...}
Mavlink1:Protocol = {MavlinkLib, UDP, UART, ...}
commands:Channel = {Mavlink1, WiFi}
WiFi:Platform = {IEEE 802.11n, WPA2, UDP, TCP/IP, ...}
Radio:Platform = {UART}
mavros_node = {. . .} . . .
```



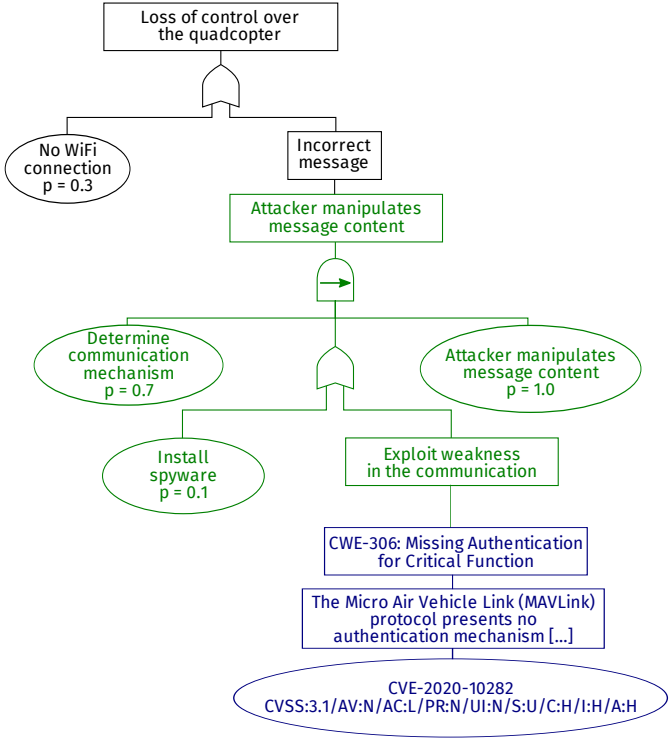
Fault Tree → AFT Fragments → Attack Fault Tree

Safety & Security – Attack Fault Tree Analysis



Fault Tree → AFT Fragments → Attack Fault Tree → Galileo Input Format

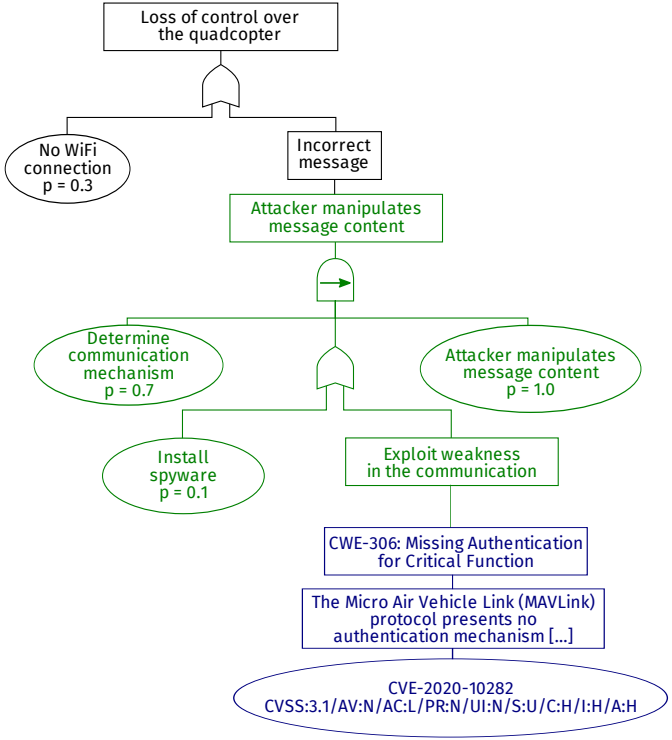
Safety & Security – Attack Fault Tree Analysis

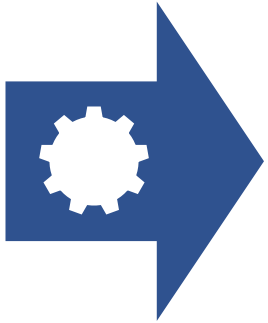



```
toplevel Loss_of_control_over_the_quadcopter;

No_Wifi_connection lambda=0.3;
Determine_communication_mechanism lambda=0.0;
Install_spyware lambda=0.000051;
Exploit_weakness_in_the_communication lambda=0.000432;
Attacker_manipulates_message_content lambda=0.0000571;

Exploit
  or Install_spyware
     Exploit_weakness_in_the_communication;

Incorrect_message
  and Determine_communication_mechanism
      Exploit
      Attacker_manipulates_message_content;

Incorrect_message_SEQ
  seq Determine_communication_mechanism
      Exploit
      Attacker_manipulates_message_content;

Loss_of_control_over_the_quadcopter
  or No_Wifi_connection
     Incorrect_message;
```

Fault Tree → AFT Fragments → Attack Fault Tree → Galileo Input Format → Model Checker

Safety & Security – Attack Fault Tree Analysis

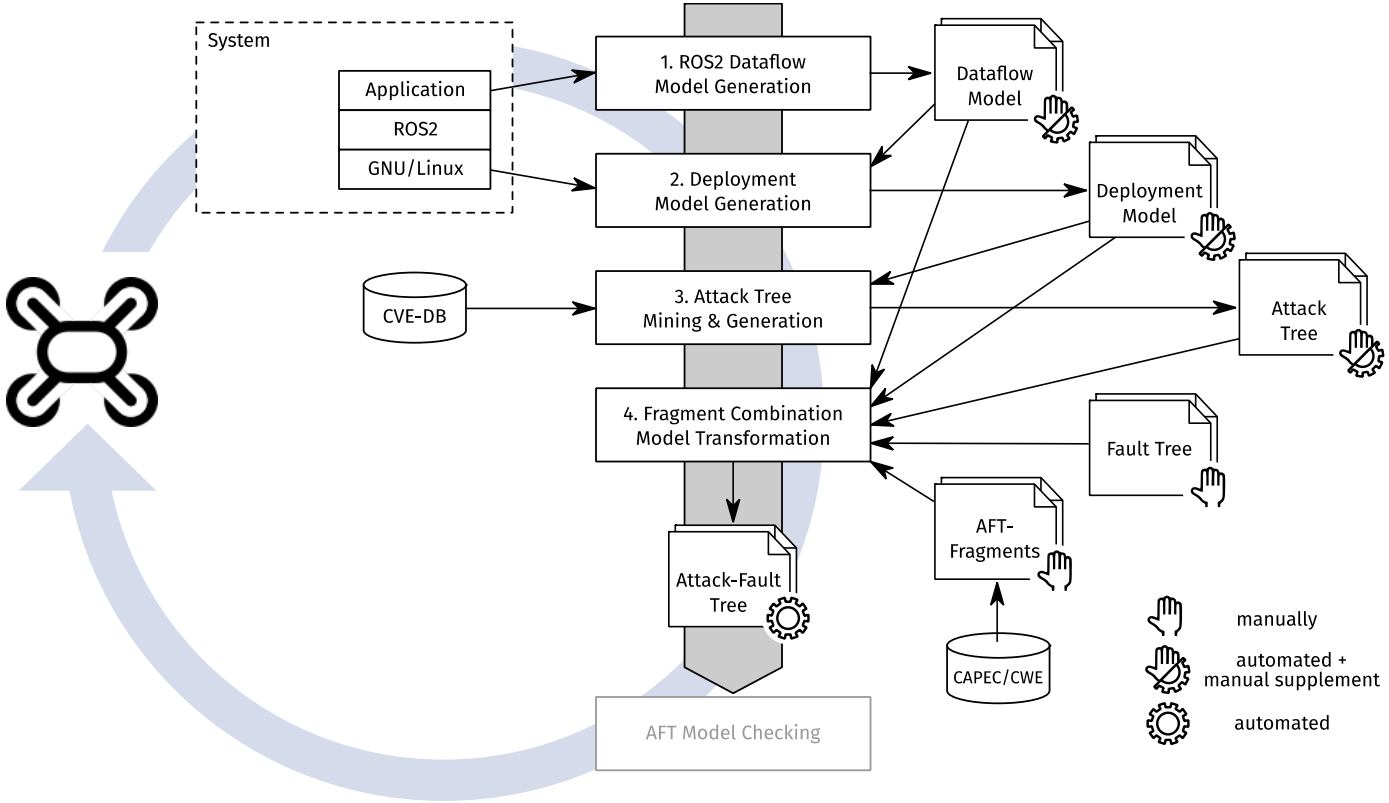




```
toplevel Loss_of_control_over_the_quadcopter;

No_Wifi_connection lambda=0.3;
Determine_communication_mechanism lambda=0.0;
Install_spyware lambda=0.000051;
Exploit_weakness_in_the_communication lambda=0.000432;
Attacker_manipulates_message_content lambda=0.0000571;

Exploit
  or Install_spyware
     Exploit_weakness_in_the_communication;

Incorrect_message
  and Determine_communication_mechanism
      Exploit
      Attacker_manipulates_message_content;

Incorrect_message_SEQ
  seq Determine_communication_mechanism
      Exploit
      Attacker_manipulates_message_content;

Loss_of_control_over_the_quadcopter
  or No_Wifi_connection
     Incorrect_message;
```

Fault Tree → AFT Fragments → Attack Fault Tree → Galileo Input Format → Model Checker

Safety & Security – Attack Fault Tree Analysis



Fault Tree → AFT Fragments → Attack Fault Tree → Galileo Input Format → Model Checker